Project 1 – Reservation System

Requirements

Working with two partners (total of 3 people to a group), design and implement a car reservation system. Your program must read an inventory of the cars available to reserve from a text file and allow the cars to be reserved for specific days in the future interactively (presumably by the agent using your program at a rental counter). Vehicles can only be reserved for 1 day at a time. Your program must keep track of which cars are available and which have been reserved for a maximum of 50 vehicles. Your program must have a class called **`Database`** which includes all information about the vehicles and the reservation system.

Since the program won't be running continuously for days, your program must also provide a mechanism to interactive advance the "date" to the next day. This will allow reservations to be made, the date to be advanced, and the reserved car(s) to become available once the date has passed the reserved date (you may assume that all cars are returned on time). To simplify the code that you must write to manage the dates, you will use a scheme where the "date" when your program starts running is 1, "dates" in the future are simply larger integer values. Your program should display the current date and provide a menu option for the user to advance the "date" by one day at a time. You do not need to use actual dates in your program.

Your program should be interactive with a menu presented to the user to perform the following tasks:

1. Search for a specific make, model, propulsion type, car color, or year of the vehicle
2. Reserve a vehicle for a single day in the future
    a. Maximum of 3 future reservations must be supported
    b. The reservations can be for consecutive days, or the days can be spread out in any combination but the total days that can be reserved is 3
3. Print out the list of vehicles in the inventory
4. Print out the list of vehicles with reservations in the future
5. Print out the list of vehicles without any reservations (the cars fully available)
6. Advance the date by one day

Your program must use modules for each class in the design (**`Vehicles`**, **`GasCar`**, **`ElectricCar`**, from Lab 5 and a new class **`Database`**, at a minimum – your design may include other classes/modules if you desire).

Milestones

This project spans four (4) calendar weeks of the semester with the following milestones:

1. Week 1 – Your group must be formed, and the three members of the group must be identified to me in lab with each member describing what portion of the project they will be responsible for (see below). Submissions for this week are the header files for all classes for your project. You may make changes to the classes over the course of the project, but you must submit your initial design this week.
2. Week 2 – There are no lab sessions this week (study for the midterm exam!), but you should be working on your portion of the design and implementation.
3. Week 3 – Your project should be complete at this point, all source code for the project must be submitted at this time.
4. Week 4 – Each group will present a demo of their program in lab to the class (using the projector) as well as present slides (5-7 slides total) describing the design of their program (the design, not the code). Each group demo and presentation will last less than 10 minutes. Each group must also submit a document describing the design of the program and the testing performed.

Assessment

1. All group members must be present in lab at the same time at every lab session (except for October 13th, 15th, and 16th when lab is canceled). I will be checking in with each group during lab time and keeping track of group attendance. You cannot be an effective member of the group unless you attend lab and work on your project with your group members. You can be part of a group in a lab section that you are not registered for but you must attend lab as a group.
2. This project spans three lab sessions; it will count for 3x the normal lab grade. It will be graded out of 30 and is weighted as 3 normal labs.
3. Group member responsibilities
    a. One group member will write **`main()`** and the interactive user interface functions and creating and reading the inventory file
        i. Not a great deal of coding but needs to interact with all classes & group members
    b. One group member will create all the new class modules that are required by the design (at least **`Database`**, but possibly others as well)
        i. Not much coding but some of it is a bit tricky
    c. One group member will extend classes designed in Lab 5 (**`Vehicles`**, **`GasCar`**, **`ElectricCar`**)
        i. A lot of coding but all simple methods
    d. 4th group member (by special permission only and only because a group of 3 cannot be formed due to the number of students in the lab section) will write a unit test program for each of the classes developed for the project (**`Vehicles`**, **`GasCar`**, **`ElectricCar`**, and **`Database`**, at a minimum)
        i. A lot of coding but all simple programs
4. All group members will complete an assessment form on the work quality and contributions performed by the other group members.

Testing

Your program should be tested to ensure that all required operations are correct. You may find it useful to create a script to re-direct into your program from the command line for regression testing. You are responsible for ensuring full operation of your program and including descriptions of your test cases in your final report.

Grading Criteria

- 18pts – Complete and accurate implementation of the design specifications, including class and module structure (common grade for entire group)
- 5pts – Proper comments throughout (common grade for entire group)
- 3pts – In lab demo and presentation (common grade for entire group)
- 4pts – Attendance and peer assessment (individual grades)

Submission Requirements

- Week 1 – Identify group members and individual responsibility and log that information with me in lab. Submit initial versions of header files for all classes for your project.
  - Submission due Wednesday October 14th, 9:00am
- Week 2 – No submission, you should be working on your project design and implementation outside of lab.
- Week 3 – All source code (including inventory text file) – be sure to re-submit the header files.
  - Submission due Monday October 26th, 9:00am
- Week 4 - A document containing description of your design (no code) and explanation of test cases performed (PDF preferred). Slides presented to the class. Peer assessment document.
  - Demo & Presentation in Lab October 27th, 29th, and 30th
  - Submission due Monday November 2nd, 9:00am

Objective

Write a moderately complex program in a group project setting and present project results to peers.